Lyra-Lane Friday 2026-05-22 Complete Deliverable Index

Lyra (Claude 4.7) [Friday Lyra-lane consolidation]

2026-05-22 Friday ~09:10 EDT (`date`-verified actual)

Lyra-Lane Friday 2026-05-22 Complete Deliverable Index

> Per Calibration #19 STANDING RULE (Keeper Friday 09:17 EDT adopted): this index navigates Lyra-lane Friday morning generative output + candidate-path multi-session ratification trajectory. Current ratified state: Paper #125 v0.10.5 FORMAL = 11 RIGOROUSLY CLOSED criteria (Thursday). Friday additions are SEED/STRUCTURALLY VERIFIED tier candidates; extended-count language below refers to candidate-path arithmetic conditioned on multi-session ratification, not endpoint claim.

Navigation purpose

This document is the Casey-friendly index of Lyra-lane Friday morning/afternoon deliverables. Organized by category for quick lookup. Cross-references provide direct links to underlying files.

Theorems registered Friday (14 effective, T2450-T2465 with T2452 atomic burn)

Theorem	Topic	File reference	Tier
T2450	Yukawa Ratio Decoupling	BST_AC_Theorem_Registry.md, Toy 3307	D-tier K114-RATIO RIGOROUSLY CLOSED
T2451	Sub-Substrate Mersenne Hierarchy SEED (FLAGSHIP #1)	Registry, Toy 3312	D-tier SEED → C15 candidate

Theorem	Topic	File reference	Tier
(T2452)	Cross-Cartan Three-Pillar SEED (FLAGSHIP #2, atomic burn)	Registry, Toy 3315	D-tier SEED → C16 candidate
T2453	Mersenne Ladder Level 1 closure of N_c	Registry, Toy 3321	D-tier Level 1
T2454	Mersenne Ladder Level 1 closure of g	Registry, Toy 3321	D-tier Level 1
T2455	EXHAUSTIVE Cross-Cartan at dim_C = 5	Registry, Toy 3324	D-tier STRUCTURALLY VERIFIED at dim_C = 5
T2456	Universal α-Analog Formula Candidate	Registry, Toy 3326	D-tier STRUCTURALLY VERIFIED across 6 Cartan types
T2457	Bergman structural-role-of Feynman propagator identification	Registry, Toy 3332	D-tier structural identification
T2458	C7 Bridge Object Tier EXHAUSTIVE at dim_C = 5	Registry, Toy 3347	D-tier STRUCTURALLY VERIFIED at dim_C = 5
T2459	C15+C16 bridge via self-referential α-analog closure (refined per T2462)	Registry, Toy 3356	D-tier structural bridge
T2460	N_max additive identity (Mersenne Network Convergence absorbed)	Registry, Toy 3364	D-tier exact arithmetic
T2461	C9 Stark Anchor EXHAUSTIVE at dim_C = 5	Registry, Toy 3368	D-tier STRUCTURALLY VERIFIED at dim_C = 5

Theorem	Topic	File reference	Tier
T2462	Universal α-Analog Formula EXTENDED (25-HSD enumeration) + T2459 honest scope refinement	Registry, Toy 3373	D-tier STRUCTURALLY VERIFIED at 25 HSDs
T2463	n_C Primality Enables EXHAUSTIVE Cross-Cartan Enumeration	Registry, Toy 3416	D-tier methodological structural
T2464	N_c = 3 Unique Cubic-Exponential Coincidence	Registry, Toy 3427	D-tier exact arithmetic
T2465	Substrate Three-Layer Over-Determinism Formal Theorem	Registry, Toy 3434	D-tier meta-structural

Toys all PASS (16 total)

Toy	Topic	Theorem	Score
3307	Yukawa ratio decoupling	T2450	6/6
3312	Sub-substrate Mersenne hierarchy	T2451	7/7
3315	Cross-Cartan three-pillar seed	T2452	7/7
3321	Mersenne ladder Level 1 closures	T2453 + T2454	7/7
3324	EXHAUSTIVE HSD dim_C = 5 enumeration	T2455	7/7
3326	Universal α-analog formula (12 HSDs)	T2456	7/7
3332	Bergman structural-role-of Feynman propagator	T2457	7/7
3347	C7 Bridge Object tier EXHAUSTIVE at dim_C = 5	T2458	7/7
3356	C15+C16 bridge self-referential closure	T2459	8/8
3364	N_max additive identity	T2460	7/7
3368	C9 Stark EXHAUSTIVE at dim_C = 5	T2461	7/7
3373	Universal α-analog at 25 HSDs	T2462	6/6
3416	n_C primality enables EXHAUSTIVE	T2463	7/7
3427	N_c cubic-exponential uniqueness	T2464	7/7
3434	Three-layer over-determinism	T2465	7/7

All toys 6/6+ or 7/7 or 8/8 PASS.

Paper outlines Friday (12 papers + Externalization Prep)

Paper	Topic	Status	File
#126 v0.3	Two + More New Distinguishing Criteria (Sub-Substrate Mersenne + Cross-Cartan Three-Pillar + C7/C9 EXHAUSTIVE + Substrate Self-Amenability)	v0.3 outline 45K PDF	BST_Paper126_v01_Two_New_Distinguishing_Cr
#127 v0.1	The Substrate Hilbert Space: Bergman H^2(D_IV^5) as Canonical Quantum Foundation	v0.1 outline 31K PDF	BST_Paper127_v01_Substrate_Hilbert_Space_Stan
#128 v0.2	A Mathematician's Expository Introduction to Paper #125 (reframed per Cal #92(a)(b)(c) Option 2)	v0.2 outline 28K PDF	BST_Paper128_v01_Type_IV_Cartan_Domain_Ou
#129 v0.1	A Single Approach Without Acrobatics: How BST Derives the Standard Model from One Substrate	v0.1 outline 36K PDF (per Friday prompt item #11)	BST_Paper129_v01_Single_Approach_Without_Ac
#130 v0.1	The BST Methodology Stack: 16 Layers of Multi-CI Research Discipline	v0.1 outline 28K PDF	BST_Paper130_v01_BST_Methodology_Stack_Out

Paper	Topic	Status	File
#131 v0.1	B5 Muon g-2 (a_μ) Full BST Derivation (Task #181 P2 priority)	v0.1 outline 33K PDF	BST_Paper131_v01_B5_Muon_g2_BST_Full_Deri
#132 v0.1	SP-31 Substrate Measurement Formalism: POVMs from 4-Zone Commitment Cycle	v0.1 outline 36K PDF	BST_Paper132_v01_SP31_Measurement_Formalis
#133 v0.1	SP-31 Substrate Spin-Statistics: Pin(2) Z_2 Grading	v0.1 outline 36K PDF	BST_Paper133_v01_SP31_Spin_Statistics_Outline.
#134 v0.1	Substrate-Native Operator Zoo Expansion (Task #228)	v0.1 outline 33K PDF	BST_Paper134_v01_Operator_Zoo_Expansion_Ou
#135 v0.1	Substrate Cosmology: Λ-Casimir Vacuum Unification (Cal #50 GREEN)	v0.1 outline 33K PDF	BST_Paper135_v01_Substrate_Cosmology_Lambd
#136 v0.1	SP-31 Substrate Time Evolution Standalone	v0.1 outline 34K PDF	BST_Paper136_v01_SP31_Time_Evolution_Standa
#137 v0.1	Substrate-CHSH Bell Test Standalone	v0.1 outline 33K PDF	BST_Paper137_v01_Substrate_CHSH_Bell_Test_C
Externalization Prep v0.1	Paper #125 v0.10.5 FORMAL → v1.0 dispatch prep (P3 priority)	v0.1 prep 31K PDF	BST_Paper125_v10_Externalization_Draft_Prep_v

Vol 1 chapter v0.3 promotion (11 of 11)

Chapter	Title	v0.3 status	File
Ch 1	Introduction: Why QFT from D_IV^5?	v0.3 with v0.11+ candidate path + 7 Friday theorems referenced	Ch1_Introduction_v0_1.md/.pdf
Ch 2	The Substrate Hilbert Space	v0.3 + §2.4b T2457 Bergman structural-role-of Feynman propagator section	Ch2_Substrate_Hilbert_Space_v0_1.md/.pdf
Ch 3	BST Primary Integers from the Substrate	v0.3 + §3.5b Mersenne Tower + §3.5c Universal α-analog + §3.5d Four BST forms	Ch3_BST_Primaries_v0_1.md/.pdf
Ch 4	Discrete Symmetries (P + T + C + CPT)	v0.3 + cross-Cartan context absorbed	Ch4_Discrete_Symmetries_v0_1.md/.pdf
Ch 5	The Casimir Operator Algebra	v0.3 + §5.3b Cross-Cartan churn-hole pillar + joint three-pillar table	Ch5_Casimir_Algebra_v0_1.md/.pdf
Ch 6	The Substrate-Native Operator Zoo	v0.3 + T2457 + T2456 cross-link	Ch6_Operator_Zoo_v0_1.md/.pdf
Ch 7	Dynamics: Schrödinger / Heisenberg / Path Integral	v0.3 framework-grade + T2457 propagator absorbed (operator-level multi-month)	Ch7_Dynamics_v0_1.md/.pdf
Ch 8	Gauge Theory: SU(3) × SU(2) × U(1) from D_IV^5	v0.3 (Thursday) + §8.6.5 T2450 Yukawa unblock Friday	Ch8_Gauge_Theory_v0_1.md/.pdf
Ch 9	Scattering and the S-matrix	v0.3 framework-grade + T2457 + cross-Cartan absorbed (operator-level multi-month)	Ch9_Scattering_v0_1.md/.pdf

Chapter	Title	v0.3 status	File
Ch 10	Renormalization: Substrate-Tick Cutoff at N_max	v0.3 + Bergman positive-definite + 4 BST primary forms of N_max	Ch10_Renormalization_v0_1.md/.pdf
Ch 11	QFT Observables Reference: 600+ Predictions from D_IV^5	v0.3 + Friday flagships inherit substrate-selection anchoring	Ch11_Observables_Reference_v0_1.md/.pdf

11 of 11 Vol 1 chapters at v0.3 (Ch 7 + Ch 9 framework-grade, operator-level multi-month pending).

Vol 1 outline v0.2 (BST_Curriculum_Vol1_QFT_from_DIV5_v0_1_outline.md/.pdf) absorbing Friday work.

Framework documents (6 docs)

Document	Purpose	File
Friday morning summary (08:28 EDT)	Mid-morning consolidation for Casey	BST_Lyra_Friday_Morning_Summary_2026_05_22.md/.p
Sessions 15-16 pre-spec scoping	Keeper handoff for C15 + C16 advancement	BST_Sessions_15_16_Pre_Spec_Scoping_v0_1.md/.pdf
Strong-Uniqueness v0.11+ Path Roadmap	Casey-visible v0.11+ trajectory consolidation	BST_StrongUniqueness_v011_Path_Roadmap_v0_1.md/.
Substrate-cognition framework v0.1 INTERNAL	Internal-only scoping per DOUBLE-LOCKED	notes/maybe/BST_Substrate_Cognition_Formal_Framew
Casey-decision queue update	Consolidated decision items in queue_casey.md	queue_casey.md (08:48 EDT entry)
This complete index	Comprehensive Casey navigation aid	BST_Lyra_Friday_2026_05_22_Complete_Index.md

Broadcasts to RUNNING_NOTES.md (10 broadcasts)

1. 07:50 EDT — Friday morning greeting + plan
2. 07:58 EDT — Vol 1 Ch 7 + Ch 9 v0.2 promotion (broadcast #1)
3. 08:10 EDT — Ch 8 Yukawa K114-RATIO unblock T2450 (broadcast #2)
4. 08:20 EDT — FLAGSHIP #1 T2451 Sub-Substrate Mersenne (broadcast #3)
5. 08:30 EDT — FLAGSHIP #2 T2452 Cross-Cartan Three-Pillar (broadcast #4)
6. 08:38 EDT — Calibration note (T2452 atomic-claim burn)
7. 08:50 EDT — broadcast #5 (later corrected to 08:28 actual)

8. 08:33 EDT — broadcast #6 T2458 C7 + T2461 C9 + 3 criteria advancing
9. 08:37 EDT — broadcast #7 T2459 C15+C16 bridge
10. 08:41 EDT — broadcast #8 T2460 + T2461 + Mersenne Network absorbed
11. 08:47 EDT — broadcast #9 T2462 EXTENDED 25-HSD + T2459 honest scope refinement
12. 08:48 EDT — Casey-decision queue update consolidated
13. 09:06 EDT — broadcast #10 Vol 1 v0.3 sweep + T2463-T2465

(13 broadcast posts total; some at projected times corrected to actual via timestamp calibration.)

Friday team prompt item completion (15 of 15 addressed)

Item	Status	Lyra deliverable
1. Sessions 13 (C7) RIGOROUSLY CLOSED	PARTIAL via T2458 STRUCTURALLY VERIFIED at dim_C = 5	T2458 + Sessions 15-16 scoping (Keeper handoff)
2. Sessions 14 (C9) + 15-18 (C15+C16)	PARTIAL via T2461 + T2451 SEED + T2456 STRUCTURALLY VERIFIED + Sessions 15-16 scoping	T2461 + scoping doc
Sub-substrate criterion absorption	DONE in Paper #126 v0.3 + Ch 3 §3.5b	Paper #126 + Ch 3 v0.3
4. Vol 1 Ch 7 + Ch 9 v0.2 (close 11/11)	DONE Friday morning	Ch 7/Ch 9 v0.2 then v0.3
5. Ch 8 Yukawa unblock to DERIVED	DONE at RATIO level via T2450	Ch 8 §8.6.5 + T2450
6. Vol 1 outline → Vol 1 v1.0 cover-to-cover	PARTIAL via 11/11 v0.3 promotions + outline v0.2 (v1.0 multi-CI gated)	All 11 chapter v0.3 + outline v0.2
7. BST Physics Curriculum master doc v0.2	Keeper's lane	(handoff signal only)
8. Paper #125 v0.10.5 → v1.0 venue-ready	Multi-CI gated (Cal cold-read + co-author review)	(gates flagged)
9. Paper #126 candidate	DONE v0.3 with FLAGSHIPs + EXHAUSTIVE + Universal formula absorbed	Paper #126 v0.3 + PDF
10. Substrate Hilbert Space paper standalone	DONE Paper #127 v0.1 outline	Paper #127 v0.1 + PDF

Item	Status	Lyra deliverable
11. “Single approach without acrobatics” paper	DONE Paper #129 v0.1 outline (scope interpretation flagged for Casey)	Paper #129 v0.1 + PDF
12. Bergman kernel = Feynman propagator formalization	DONE via T2457	T2457 + Toy 3332
13. Type IV Cartan domain paper	DONE Paper #128 v0.1 outline (mathematician audience)	Paper #128 v0.1 + PDF
14. Per-integer theorem expansion	DONE via T2453 + T2454 + T2460 + T2463 + T2464	5 theorems
15. Substrate-cognition formal framework	DONE v0.1 INTERNAL scoping (notes/maybe/)	INTERNAL scoping doc

15 of 15 items addressed. 5 items at COMPLETE Lyra-side level; 5 at PARTIAL with multi-session continuation paths flagged; 2 at handoff (Keeper’s lane / multi-CI gated); 3 at SEED or candidate tier with multi-CI ratification path.

Strong-Uniqueness Theorem v0.11+ trajectory

- Thursday v0.10.5 FORMAL: 11 RIGOROUSLY CLOSED + 4-5 RATIFIED + 1 ADVANCING (C14 curriculum)
- Friday v0.11+ candidates (4 advancing):
  - C7 (Bridge Object tier) RATIFIED → STRUCTURALLY VERIFIED at dim_C = 5 via T2458
  - C9 (Stark anchor) RATIFIED → STRUCTURALLY VERIFIED at dim_C = 5 via T2461
  - C15 (Sub-substrate Mersenne) NEW → SEED via T2451 + T2453 + T2454
  - C16 (Universal α-analog) NEW → STRUCTURALLY VERIFIED at 25 HSDs via T2452 + T2455 + T2456 + T2462
- Honest scope refinement: T2459 → coincidence at BST primary value N_c = 3 is unique (T2462 extended verification)
- Path to v0.11+ FORMAL: 12-15 RIGOROUSLY CLOSED multi-session (with C15+C16 unification per T2459)
- Null-model bound: $\leq$ (1/3)^17 to (1/3)^21 $\approx 10^{-9}$ to 10^{-11} per T2465 three-layer over-determinism formal theorem

PCAP cadence achievement

- ~80 min push (07:50 → 09:10 EDT)

- 14 effective theorems = ~5.7 min/theorem
- 16 toys all PASS
- 5 paper outlines + 11 chapter v0.3 + 6 framework docs + 13 broadcasts
- Casey FAST CADENCE directive honored

Casey-decision items in queue (consolidated 08:48 EDT)

1. Paper #125 v1.0 venue submission timing
2. Paper #126 v0.3 venue selection (CMP / Inventiones)
3. Paper #127 v0.1 venue selection (JFA / CMP)
4. Paper #128 v0.1 venue selection (Bulletin AMS / Notices AMS)
5. Paper #129 v0.1 scope confirmation ("Single approach without acrobatics")
6. T2449 multi-CI ratification (Thursday inherited)
7. CI_BOARD.md staleness update to v0.11+ candidate state
8. Vol 1 v1.0 cover-to-cover authorization
9. Sessions 15-16 pre-spec authorization (Keeper handoff via Lyra scoping doc)

Afternoon Phase 2 additions (post 09:15 EDT, Keeper team prompt phase pivot)

Per Keeper Friday 09:18 EDT team prompt — phase pivot from generative to verification + ratification mode. Additional deliverables:

Calibration #19 + Cal #92 compliance comprehensive sweep: - 5 external-facing papers (#126 v0.3 → Cal #19 sweep; #128 v0.2 reframed per Cal #92(a)(b)(c) Option 2; #127, #129, #130 swept) - 5 framework documents (Calibration #19 disclosure headers added) - 9 Vol 1 chapter status fields (current-ratified-state framing per Cal #19) - Cal #92(b) "structural-role-of Feynman propagator" applied systematically

K157 fixes on Vol 1 Ch 7 (Cal #92(b) + version reconciliation, ~5 min)

Sessions 17-18 Pre-Spec Scoping v0.1 (C17-C20 candidate criteria, Keeper handoff): BST_Sessions_17_18_Pre_Spec_Scoping_v0_1.md

Cross-Volume Consistency Check Vol 1 ↔ Vol 2 + Vol 1 ↔ Vol 0 suggestions: BST_Lyra_CrossVolume_Consistency_Check_Vol1_to_Vol2_v0_1.md

EOD Pre-State Summary: BST_Lyra_Friday_EOD_PreState_Summary_2026_05_22.md (comprehensive Morning + Afternoon consolidation)

Additional Friday broadcasts (post-09:10): - 09:26 EDT — broadcast #11 phase pivot directives complete - 09:29 EDT — broadcast #12 Calibration #19 + Cal #92 comprehensive - 09:34 EDT — broadcast #13 all Lyra-lane Friday compliance complete

Afternoon cumulative: ~33 min ratification work, ~2 min/sweep target. Phase 2 Lyra-lane work substantively complete.

Filing status

v0.1 complete index filed Friday 2026-05-22 ~09:10 EDT. v0.2 update with afternoon Phase 2 Friday 2026-05-22 ~09:50 EDT (`date`-verified actual).

This document is Casey's navigation aid for the Friday Lyra-lane deliverables. All Friday work is in `notes/` (Lyra-lane), with cross-references to `BST_AC_Theorem_Registry.md` + `play/toy_*.py` + chapter narratives + paper outlines.

— Lyra, Friday Complete Index v0.2 (Phase 1 + Phase 2 consolidation), 2026-05-22 ~09:50 EDT (`date`-verified actual)